

H2O Extreme-Move Path (efficient momentum)

Continuation after a large, smooth 30-minute move on NQ futures: research report for the trading decision (cohort member #1 Cdf2b568ee1f8a182)

Prepared for the trading decision committee · 23 September 2026 · data through 10 August 2026 · all figures per 1 NQ contract unless stated

Recommendation: CONDITIONAL. Do not allocate full size now; start a small, fixed-risk pilot alongside the true-forward record. This is the best-evidenced strategy in the campaign. It passed every preregistered stage including the untouched 2023–25 holdout (+\$120.6k, PF 1.21, every year positive). But it lost –\$52.6k in 2026, and it is the cohort's most correlated member. Scale up only after the forward gate on the last page is met.

Executive summary

- **What it is.** When NQ's 30-minute move exceeds 2 standard deviations *and* the path was smooth (normalised path efficiency ≥ 2.4 , a steady move rather than a jump), trade in the direction of the move at the next bar. Stop is $1.5 \times \text{ATR}$, there is no target, and the position is flat at 16:00 New York. About 260 trades a year, long and short.
- **Evidence ladder.** Discovery 2010–18: +\$52.2k, 2,096 trades, PF 1.17. Confirmation 2019–22: +\$149.1k, 1,045 trades, PF 1.26, t 2.56, null p 0.039, STRONG. **Holdout 2023–25 (never seen): +\$120.6k, 795 trades, PF 1.21, +149R, all three years positive, SUPPORTED.** The 2023–25 average trade (+0.19R) was higher than in the periods used to select it.
- **2024–2025 alone:** 526 trades, +\$96.9k, PF 1.23, win rate 38%, +104.6R, max drawdown \$42.9k.
- **2026 was bad in dollars, roughly flat in R.** Jan–Aug 2026: 167 trades, –\$52.6k, PF 0.77, max drawdown \$62.6k, preregistered FAILED. In R it was only –4.3R (–0.03R per trade). NQ's volatility, and so the dollar size of each $1.5 \times \text{ATR}$ stop, roughly doubled (about \$2,275 per trade vs \$1,050 in 2019–22). Resampling its own 2019–25 trades gives a result this weak about 11% of the time.
- **It is a behaviour, not a lucky parameter.** All 12 continuation/high-efficiency neighbours had PF > 1 in discovery, confirmation *and* the holdout. 27 of 29 stop/exit variants were profitable in the holdout. Fading the same moves (reversal mode) mostly lost money. The flip side: 11 of the 12 neighbours and 28 of 29 management variants also lost in 2026, so the 2026 weakness hit the whole behaviour.
- **Main risks.** (1) A fat right tail: after 2022 the 10 best trades exceed 100% of net profit, so missing a few big days erases the year. (2) Correlation 0.64 with the rest of the cohort (it overlaps the H41/H43 momentum strategies), so it adds less diversification than its standalone numbers suggest. (3) Dollar risk per contract is now about $2 \times$ its 2019–22 level.

Headline results by research stage

Strategy	Period	Trades	Net P&L	PF	Win %	Net R	Avg R	Max DD	Verdict
H2O	Discovery 2010–18	2,096	\$52,191	1.17	37.3%	+127.5	+0.061	\$9,267	PRIMARY
H2O	Confirmation 2019–22	1,045	\$149,075	1.26	40.4%	+120.6	+0.115	\$25,869	STRONG
H2O	Holdout 2023–25	795	\$120,595	1.21	38.5%	+148.8	+0.187	\$42,886	SUPPORTED
H2O	2026 audit (Jan–Aug)	167	–\$52,593	0.77	32.9%	–4.3	–0.025	\$62,565	FAILED
H2O	2024–2025 only	526	\$96,876	1.23	38.0%	+104.6	+0.199	\$42,886	–
H2O	All post-freeze 2023–26	962	\$68,002	1.08	37.5%	+144.6	+0.150	\$62,565	–

Stages before the freeze (discovery, confirmation) were used to select the strategy; 2023 onward was never seen before the cohort was frozen on 22 Sep 2026 23:21 UTC. Verdicts are the campaign's preregistered classifications. R = P&L divided by the trade's own initial risk.

1. How this strategy was found and tested

The strategy came out of the QuantLab V3 research campaign: 53 preregistered hypothesis families (33,077 specifications) were searched on 2010–2018 NQ/ES 1-minute data. Survivors were frozen and re-tested unchanged on 2019–2022 against 100 synthetic ‘no-edge’ market worlds, clustered into distinct behaviours, and 25 were selected into a frozen cohort. The cohort was then run once on the untouched 2023–2025 holdout, and once on 2026 data to 10 Aug 2026. Nothing was changed after any later stage was seen; that is what makes the post-2022 numbers honest out-of-sample evidence.

Stage	Dates	Role	What could change
Discovery	2010-06-07 → 2018-12-31	search (in-sample)	everything: rules, grid, management
Confirmation	2019 → 2022	out-of-sample test vs null worlds	only which frozen candidates are kept
Final cohort freeze	22 Sep 2026 23:21 UTC	spec hash f5666dac...	nothing after this point
Holdout	2023 → 2025	untouched out-of-sample	nothing (report only)
2026 audit	2026-01-01 → 2026-08-10	secondary, low power	nothing (report only)
True forward	from 2026-09-24	live-market paper record	nothing (forward/run_forward.py)

Reproducibility. For this report every trade from 2010 to 2026 was regenerated with the frozen specification, the stage-gated data loader and the campaign engine. Trade counts and net P&L match the sealed campaign records exactly in all four periods for every strategy covered here.

Why R-multiples matter here. NQ traded near 2,000 in 2010 and near 27,000 in 2026, and the stop is volatility-based, so one trade's dollar risk ranged from roughly \$50 to over \$2,000 on one contract. Dollar totals are therefore dominated by recent, high-volatility years, and a single contract is a very different bet in 2012 than in 2026. We report both dollars (the campaign's preregistered metric) and R ($\text{P\&L} \div \text{initial risk}$), which approximates a book that risks a constant amount per trade. **The official verdicts use dollars; the R view is a descriptive, post-hoc lens and cannot overturn them.**

Hypothesis family

Preregistered hypothesis (H20). Equally large moves $Z(n) \geq z$ are split by path efficiency: HIGH (normalised efficiency ≥ 2.4) or LOW (≤ 1.7). Each is tested as continuation *and* reversal. Grid: $n \in \{30, 60\}$, $z \in \{2.0, 2.5\}$, class $\in \{\text{high, low}\}$, mode $\in \{\text{continuation, reversal}\}$, both / long / short: 48 entry rules and 1,128 specifications. In discovery 80 were PRIMARY (best t 2.41, median PF 0.94).

Rationale. A large move built from many same-direction minutes (high efficiency) suggests persistent order flow: institutions working an order, or information being absorbed gradually. That tends to continue for the rest of the session. A large move made in one jump with a noisy path is more often liquidity-driven and does not persist. The design tests both sides of this. Continuation of efficient moves worked; fading them lost money in nearly every variant before 2026.

Behaviour cluster. The strategy represents the campaign's largest behaviour: cluster #43, STRONG, 191 confirmation survivors and 41 distinct entry rules across H01, H02, H03, H04, H10, H20, H27, H30 and X3. Many independently specified momentum rules found the same effect, the best protection against a single-rule fluke. The efficiency filter raised the average trade in 69% of discovery and 90% of confirmation paired tests (median +\$26 per trade).

2. H20 (#1, long + short): specification

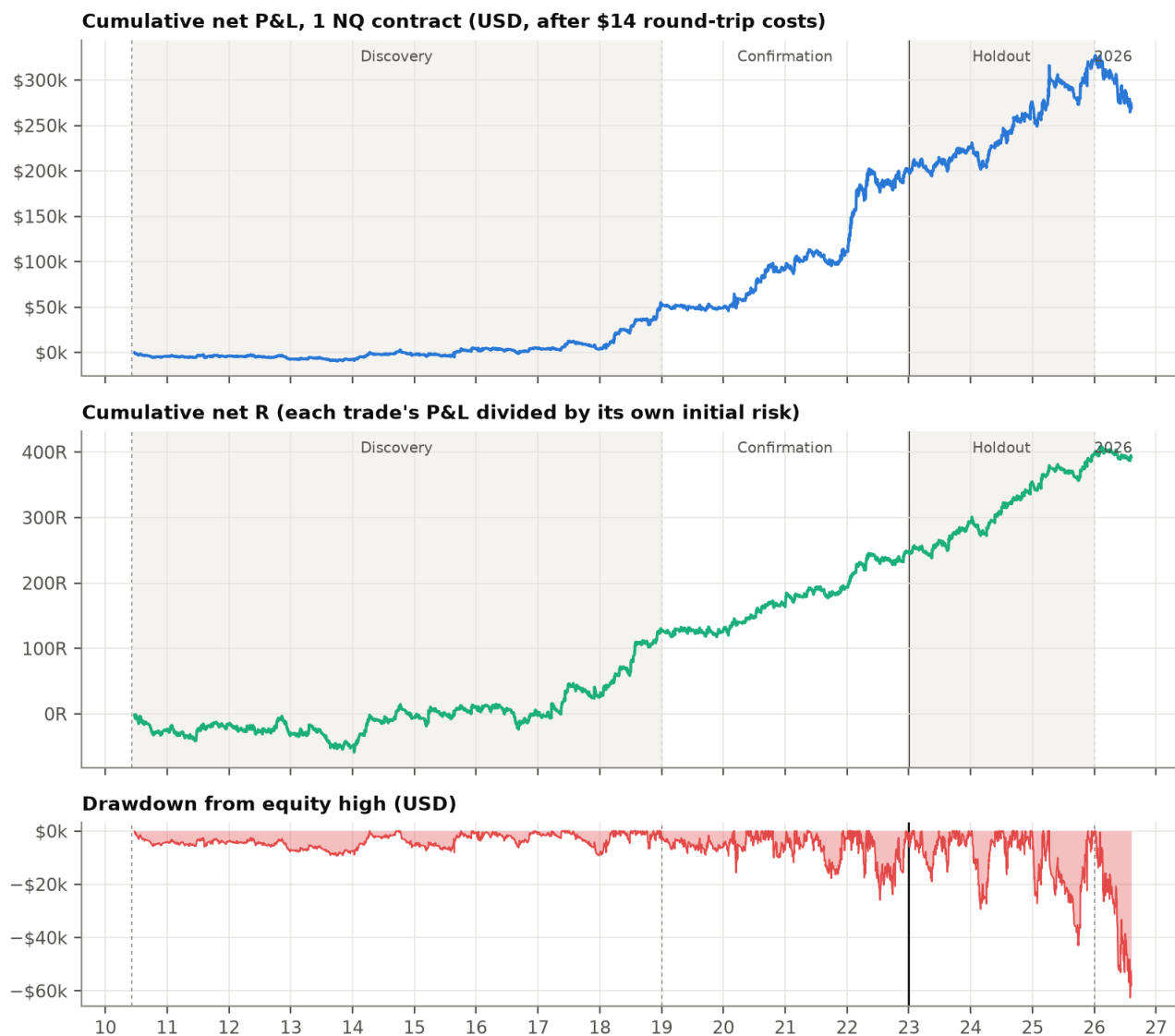
Field	Frozen value
Strategy ID	Cdf2b568ee1f8a182
Entry rule ID	Eac20f7491875358c
Family / version	v3_h20_extreme_move_path (v3.0)
Instrument	NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars
Direction	both
Mode	continuation
Long trigger	the 30-minute move first exceeds +2.0 sigma units with path efficiency $\times \sqrt{n} \geq 2.4$
Short trigger	the 30-minute move first falls below -2.0 sigma units with path efficiency $\times \sqrt{n} \geq 2.4$
Parameters	{"cls": "high", "mode": "continuation", "n": 30, "pe_hi": 2.4, "pe_lo": 1.7, "z": 2.0}
Signal timing	evaluated at the close of each 1-minute bar
Entry order	market order at the open of the next 1-minute bar (same session and contract)
Entry window	09:30-15:30 New York; one position at a time; no pyramiding
Initial stop	1.5 $\times$ ATRh from entry (ATRh = ATR(14) of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick
Target	none (100% of the position runs)
Break-even / trailing / partials	none / none / none
Time exit	flat at the close of the last bar before 16:00 New York
Same-bar ambiguity	pessimistic (stop assumed first)
Costs (baseline)	\$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26
Cost file version	costs_v2_approved_2026-09-21#a27ed5df0c04

Statistical evidence trail

Stage	Test / statistic	Result	Verdict
Discovery 2010-18	label (trades ≥ 30 , net > 0 , t ≥ 1)	2096 trades, PF 1.17, t 2.24	PRIMARY
Discovery	parameter robustness (share of grid neighbours positive)	1.00	
Discovery	sub-periods positive (of 3) / years positive (of 9)	2 / 5	
Confirmation 2019-22	t-stat of mean trade	2.56	
Confirmation	null p (zero-drift worlds) / (drift worlds) / max	0.039 / 0.020 / 0.039	$p \leq 0.05$
Confirmation	family-wise p (max-t in same frequency bucket)	0.119	not significant
Confirmation	parameter support / management support	1.00 / 0.62	
Confirmation	years positive (of 4); stress net	3; \$136,535	
Confirmation	evidence tier (preregistered)		STRONG
Behaviour cluster	cluster at p 0.50 (members / entry rules)	#43 (191 / 41)	STRONG
Holdout 2023-25	t-stat; z vs confirmation mean trade	1.73; 0.10	
Holdout	null p (100 holdout null worlds)	0.059	
Holdout	classification (preregistered)		SUPPORTED
2026 audit	t-stat; net	-1.36; -\$52,593	FAILED

Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.

H20 (#1, long + short): performance



Shaded bands mark the research stages; the solid black line is the start of data never seen before the freeze (1 Jan 2023).

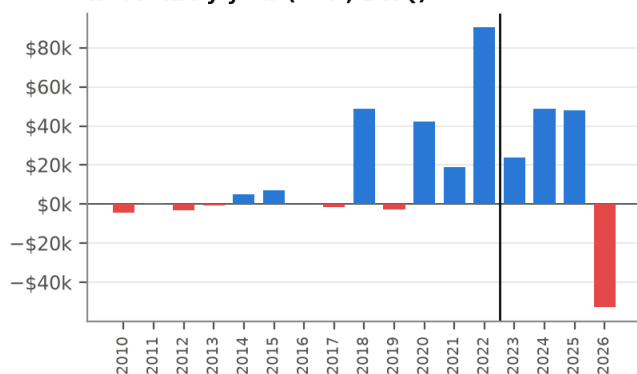
H20 (#1, long + short): full statistics by period

Metric	Discovery 2010-18	Confirm. 2019-22	Holdout 2023-25	2026 Jan-Aug	2024-25	Post-freeze 2023-26	Full 2010-26
Trades	2,096	1,045	795	167	526	962	4,103
Trades / year	245	261	265	271	263	266	253
Long share	47%	45%	49%	43%	47%	48%	47%
Net P&L (baseline \$14)	\$52,191	\$149,075	\$120,595	-\$52,593	\$96,876	\$68,002	\$269,268
Gross P&L (no costs)	\$81,535	\$163,705	\$131,725	-\$50,255	\$104,240	\$81,470	\$326,710
Net P&L, moderate \$9	\$62,671	\$154,300	\$124,570	-\$51,758	\$99,506	\$72,812	\$289,783
Net P&L, stress \$26	\$27,039	\$136,535	\$111,055	-\$54,597	\$90,564	\$56,458	\$220,032
Profit factor (net)	1.17	1.26	1.21	0.77	1.23	1.08	1.16
Profit factor (gross)	1.28	1.29	1.23	0.78	1.25	1.10	1.20
Profit factor (stress)	1.08	1.23	1.19	0.76	1.21	1.07	1.13
Win rate	37.3%	40.4%	38.5%	32.9%	38.0%	37.5%	38.1%
Average trade	\$25	\$143	\$152	-\$315	\$184	\$71	\$66
Average winner	\$458	\$1,728	\$2,272	\$3,145	\$2,607	\$2,405	\$1,250
Average loser	-\$232	-\$931	-\$1,175	-\$2,014	-\$1,302	-\$1,332	-\$664
Payoff ratio (avg W / avg L)	1.97	1.86	1.93	1.56	2.00	1.81	1.88
Expectancy (net R / trade)	+0.061	+0.115	+0.187	-0.025	+0.199	+0.150	+0.096
Total net R	+127.5	+120.6	+148.8	-4.3	+104.6	+144.6	+392.6
Median trade (R)	-1.03	-1.01	-1.00	-1.00	-1.00	-1.00	-1.01
t-stat of mean trade (USD)	2.24	2.56	1.73	-1.36	1.50	0.85	2.65
t-stat of mean trade (R)	1.51	2.22	2.57	-0.20	2.15	2.34	3.33
Sharpe, daily, annualised (USD)	0.79	1.27	1.00	-1.63	1.06	0.44	0.65
Sharpe, daily, annualised (R)	0.51	1.10	1.46	-0.25	1.50	1.21	0.82
Sortino (USD)	1.45	2.19	1.91	-2.24	2.05	0.76	1.13
Max drawdown (USD)	\$9,267	\$25,869	\$42,886	\$62,565	\$42,886	\$62,565	\$62,565
Max drawdown (R)	58.5	17.7	28.4	21.4	28.4	28.4	58.5
Longest drawdown (calendar days)	1,388	272	226	212	226	226	1,388
Calmar (annual net / max DD)	0.66	1.44	0.94	-1.36	1.13	0.30	0.27
Longest losing streak (trades)	14	10	13	12	13	13	14
Longest winning streak	6	6	5	4	5	5	6
Best trade	\$5,286	\$12,511	\$20,991	\$10,416	\$20,991	\$20,991	\$20,991
Worst trade	-\$1,754	-\$3,559	-\$6,689	-\$5,129	-\$6,689	-\$6,689	-\$6,689
Best day / worst day	\$4,692 / -\$3,606	\$12,511 / -\$8,381	\$19,191 / -\$10,938	\$10,181 / -\$10,921	\$19,191 / -\$10,938	\$19,191 / -\$10,938	\$19,191 / -\$10,938
Top-10 trades as % of net	71%	56%	112%	-	140%	205%	53%
Positive months	50% of 103	62% of 48	64% of 36	38% of 8	62% of 24	59% of 44	55% of 195
Average hold (minutes)	144	150	149	146	150	148	147
Avg initial risk (NQ points)	12.9	52.6	68.7	113.7	76.1	76.5	37.9
Avg initial risk, 1 NQ (USD) = 1R	\$257	\$1,053	\$1,373	\$2,275	\$1,521	\$1,530	\$758
Average NQ price at entry	4,082	11,460	18,578	27,219	20,792	20,078	9,712

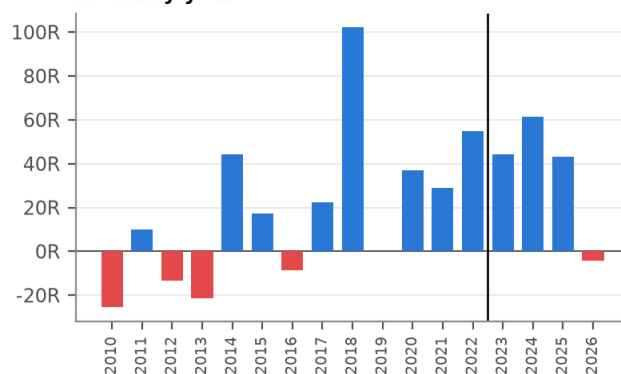
Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

H20 (#1, long + short): calendar

Net P&L by year (USD, 1 NQ)



Net R by year



Year	Period	N	Net P&L	Gross	Stress	PF	Win %	Avg trade	Net R	Avg R	Max DD	1R (USD)
2010	Disc.	118	-\$4,212	-\$2,560	-\$5,628	0.64	36%	-\$36	-25.4	-0.215	\$5,633	\$148
2011	Disc.	251	\$371	\$3,885	-\$2,641	1.01	39%	\$1	+10.0	+0.040	\$3,379	\$196
2012	Disc.	248	-\$3,057	\$415	-\$6,033	0.88	33%	-\$12	-13.2	-0.053	\$5,141	\$162
2013	Disc.	247	-\$703	\$2,755	-\$3,667	0.97	38%	-\$3	-21.4	-0.087	\$3,949	\$150
2014	Disc.	221	\$5,116	\$8,210	\$2,464	1.19	39%	\$23	+44.3	+0.200	\$6,176	\$218
2015	Disc.	247	\$7,172	\$10,630	\$4,208	1.18	38%	\$29	+17.2	+0.070	\$5,104	\$298
2016	Disc.	258	\$238	\$3,850	-\$2,858	1.01	34%	\$1	-8.5	-0.033	\$6,065	\$270
2017	Disc.	238	-\$1,452	\$1,880	-\$4,308	0.96	35%	-\$6	+22.3	+0.094	\$9,183	\$229
2018	Disc.	268	\$48,718	\$52,470	\$45,502	1.64	43%	\$182	+102.2	+0.381	\$6,676	\$559
2019	Conf.	232	-\$2,873	\$375	-\$5,657	0.95	37%	-\$12	+0.2	+0.001	\$6,748	\$454
2020	Conf.	262	\$42,407	\$46,075	\$39,263	1.29	42%	\$162	+36.9	+0.141	\$15,624	\$1,051
2021	Conf.	271	\$19,136	\$22,930	\$15,884	1.13	42%	\$71	+28.8	+0.106	\$17,739	\$1,017
2022	Conf.	280	\$90,405	\$94,325	\$87,045	1.39	40%	\$323	+54.7	+0.195	\$25,869	\$1,585
2023	Holdout	269	\$23,719	\$27,485	\$20,491	1.16	39%	\$88	+44.2	+0.164	\$18,957	\$1,083
2024	Holdout	278	\$48,978	\$52,870	\$45,642	1.27	38%	\$176	+61.3	+0.220	\$29,253	\$1,261
2025	Holdout	248	\$47,898	\$51,370	\$44,922	1.20	38%	\$193	+43.3	+0.175	\$42,886	\$1,813
2026	Audit	167	-\$52,593	-\$50,255	-\$54,597	0.77	33%	-\$315	-4.3	-0.025	\$62,565	\$2,275

Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	-2.2	-7.3	+11.0	+2.3	-6.3	+2.5	-1.8	-0.5	-0.3	-1.6	-1.2	+5.7
2020	-1.1	+11.7	+2.4	-0.2	+5.2	+3.0	+7.9	+7.2	+5.4	-2.1	-1.3	-1.2
2021	+14.6	+2.0	-0.9	-0.2	+12.9	+0.2	-3.6	-10.4	+1.3	+4.3	+6.8	+1.9
2022	+20.3	+12.1	+2.2	+14.3	+0.6	-5.3	-0.1	-0.7	-3.8	+4.7	+4.2	+6.3
2023	+8.6	-4.8	-4.3	-4.7	+13.4	+6.9	-2.4	+10.9	+5.5	+2.7	+1.8	+10.7
2024	-5.6	-13.2	+1.6	+20.6	+6.8	+4.9	+8.5	+5.5	+5.3	+10.2	+2.0	+14.5
2025	-6.8	+18.1	+1.4	+9.0	+4.1	-5.9	-3.7	-3.8	-8.5	+17.7	+14.5	+7.1
2026	-1.5	+3.5	+2.3	-5.1	-5.3	+1.4	-5.2	+5.6				

H20 (#1, long + short): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

H20 (#1, long + short): breakdowns (all periods vs post-freeze 2023-26)

Direction

Direction	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Long	1913	+0.118	1.20	\$144,498	464	+0.101	1.06	\$21,029
Short	2190	+0.076	1.13	\$124,770	498	+0.196	1.11	\$46,973

Exit reason

Exit	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
16:00 flat	1934	+1.381	16.46	\$1,836,109	463	+1.403	13.98	\$806,208
Stop-loss hit	2169	-1.050	0.00	-\$1,566,841	499	-1.012	0.00	-\$738,206

Entry hour (New York)

Entry hour	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
09:00-09:59	719	+0.364	1.47	\$118,469	173	+0.784	1.87	\$99,408
10:00-10:59	983	+0.002	1.02	\$11,418	246	-0.077	0.94	-\$13,379
11:00-11:59	800	+0.106	1.16	\$56,400	162	-0.015	0.86	-\$24,148
12:00-12:59	582	+0.066	1.13	\$29,932	143	-0.027	0.90	-\$11,907
13:00-13:59	432	+0.052	1.33	\$51,532	105	+0.136	1.37	\$25,565
14:00-14:59	410	-0.047	1.01	\$870	93	+0.278	1.05	\$3,003
15:00-15:59	177	+0.015	1.02	\$647	40	-0.147	0.46	-\$10,540

Weekday

Weekday	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Mon	690	+0.157	1.16	\$45,140	166	+0.010	0.97	-\$3,214
Tue	855	+0.138	1.15	\$50,165	179	+0.209	1.13	\$18,859
Wed	879	+0.024	1.10	\$34,654	200	+0.026	0.94	-\$11,000
Thu	903	+0.097	1.24	\$90,333	219	+0.297	1.20	\$38,069
Fri	776	+0.074	1.14	\$48,976	198	+0.179	1.15	\$25,288

H20 (#1, long + short): robustness

Parameter neighbourhood. Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
n=30, z=2.0, cls=high, mode=continuation	long	1123 / 1.19	555 / 1.36	466 / 1.16	\$51,131	84 / 0.97	−\$3,226
n=30, z=2.0, cls=high, mode=continuation	short	1197 / 1.07	625 / 1.17	448 / 1.21	\$74,258	106 / 0.76	−\$32,259
n=30, z=2.0, cls=high, mode=continuation ◀ frozen	both	2096 / 1.17	1045 / 1.26	795 / 1.21	\$120,595	167 / 0.77	−\$52,593
n=30, z=2.5, cls=high, mode=continuation	long	700 / 1.20	344 / 1.34	273 / 1.09	\$20,228	47 / 0.96	−\$2,488
n=30, z=2.5, cls=high, mode=continuation	short	855 / 1.03	447 / 1.17	341 / 1.12	\$35,051	69 / 0.82	−\$16,056
n=30, z=2.5, cls=high, mode=continuation	both	1442 / 1.12	722 / 1.25	550 / 1.11	\$48,065	104 / 0.76	−\$33,896
n=60, z=2.0, cls=high, mode=continuation	long	762 / 1.23	379 / 1.32	312 / 1.19	\$40,672	57 / 1.28	\$17,082
n=60, z=2.0, cls=high, mode=continuation	short	918 / 1.10	457 / 1.08	343 / 1.08	\$23,383	62 / 0.83	−\$11,768
n=60, z=2.0, cls=high, mode=continuation	both	1615 / 1.17	796 / 1.18	629 / 1.13	\$61,374	112 / 1.04	\$5,387
n=60, z=2.5, cls=high, mode=continuation	long	455 / 1.17	217 / 1.25	165 / 1.32	\$38,055	31 / 0.91	−\$3,219
n=60, z=2.5, cls=high, mode=continuation	short	613 / 1.12	323 / 1.06	242 / 1.04	\$9,847	42 / 0.80	−\$10,873
n=60, z=2.5, cls=high, mode=continuation	both	1039 / 1.13	516 / 1.13	398 / 1.13	\$43,848	69 / 0.82	−\$15,466
n=30, z=2.0, cls=high, mode=reversal	long	1539 / 0.90	784 / 0.87	580 / 0.89	−\$50,735	132 / 1.66	\$89,387
n=30, z=2.0, cls=high, mode=reversal	short	1317 / 0.85	635 / 0.91	529 / 0.79	−\$84,016	98 / 0.77	−\$24,052
n=30, z=2.0, cls=high, mode=reversal	both	2339 / 0.88	1141 / 0.90	893 / 0.89	−\$72,017	180 / 1.42	\$78,885
n=30, z=2.5, cls=high, mode=reversal	long	1088 / 0.91	544 / 0.84	427 / 0.97	−\$9,113	87 / 1.10	\$11,257
n=30, z=2.5, cls=high, mode=reversal	short	802 / 0.86	380 / 0.84	307 / 0.80	−\$52,693	55 / 0.79	−\$14,550
n=30, z=2.5, cls=high, mode=reversal	both	1587 / 0.87	780 / 0.80	613 / 0.96	−\$17,872	117 / 0.92	−\$12,533
n=60, z=2.0, cls=high, mode=reversal	long	1180 / 0.77	592 / 1.03	449 / 1.02	\$5,349	81 / 1.12	\$11,536
n=60, z=2.0, cls=high, mode=reversal	short	911 / 0.68	452 / 0.95	365 / 0.85	−\$43,310	65 / 0.84	−\$11,455
n=60, z=2.0, cls=high, mode=reversal	both	1838 / 0.70	888 / 0.96	694 / 0.96	−\$23,001	129 / 0.97	−\$4,771
n=60, z=2.5, cls=high, mode=reversal	long	769 / 0.82	414 / 1.10	320 / 0.94	−\$16,880	56 / 1.28	\$18,701
n=60, z=2.5, cls=high, mode=reversal	short	527 / 0.77	239 / 0.88	188 / 0.89	−\$19,352	34 / 0.95	−\$1,936
n=60, z=2.5, cls=high, mode=reversal	both	1162 / 0.76	583 / 1.00	447 / 0.91	−\$34,243	80 / 1.10	\$10,270
n=30, z=2.0, cls=low, mode=continuation	long	513 / 0.90	319 / 1.02	173 / 1.31	\$51,773	44 / 1.30	\$16,269
n=30, z=2.0, cls=low, mode=continuation	short	667 / 1.00	369 / 0.95	260 / 1.00	\$820	50 / 1.11	\$7,130
n=30, z=2.0, cls=low, mode=continuation	both	945 / 0.97	530 / 0.95	339 / 1.29	\$88,414	70 / 0.85	−\$14,285
n=30, z=2.5, cls=low, mode=continuation	long	209 / 0.89	160 / 1.30	74 / 1.53	\$41,349	22 / 1.11	\$3,137
n=30, z=2.5, cls=low, mode=continuation	short	288 / 1.06	195 / 1.12	125 / 1.04	\$6,340	23 / 1.48	\$13,058
n=30, z=2.5, cls=low, mode=continuation	both	419 / 1.00	278 / 1.26	165 / 1.64	\$95,185	34 / 0.80	−\$9,171
n=60, z=2.0, cls=low, mode=continuation	long	288 / 0.90	209 / 0.95	107 / 1.47	\$46,782	29 / 1.36	\$13,454
n=60, z=2.0, cls=low, mode=continuation	short	409 / 1.06	255 / 1.04	177 / 0.98	−\$3,428	30 / 1.26	\$9,180
n=60, z=2.0, cls=low, mode=continuation	both	611 / 1.00	384 / 0.93	248 / 1.26	\$59,743	50 / 1.37	\$21,385
n=60, z=2.5, cls=low, mode=continuation	long	118 / 1.23	91 / 1.14	46 / 1.75	\$38,921	10 / 1.37	\$6,195
n=60, z=2.5, cls=low, mode=continuation	short	188 / 1.00	137 / 1.17	70 / 1.16	\$12,015	14 / 1.24	\$3,859

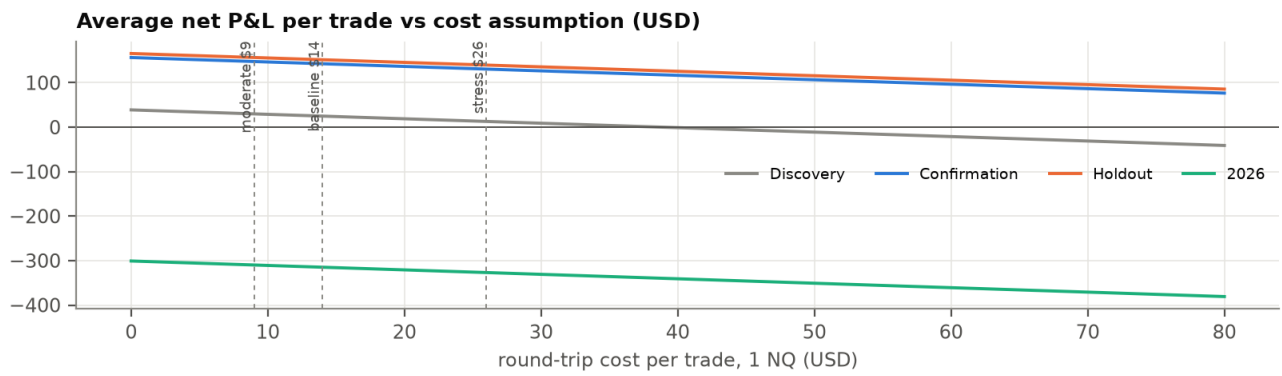
Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
n=60, z=2.5, cls=low, mode=continuation	both	272 / 1.08	201 / 1.09	108 / 1.32	\$38,538	21 / 1.27	\$7,296
n=30, z=2.0, cls=low, mode=reversal	long	890 / 0.92	488 / 1.12	344 / 0.97	-\$8,831	69 / 0.96	-\$4,726
n=30, z=2.0, cls=low, mode=reversal	short	546 / 1.15	329 / 1.07	193 / 0.81	-\$43,952	47 / 0.73	-\$21,173
n=30, z=2.0, cls=low, mode=reversal	both	1052 / 1.02	567 / 1.05	411 / 0.76	-\$105,144	90 / 0.96	-\$5,035
n=30, z=2.5, cls=low, mode=reversal	long	389 / 0.84	276 / 1.07	155 / 0.72	-\$50,745	34 / 1.06	\$3,634
n=30, z=2.5, cls=low, mode=reversal	short	233 / 1.18	165 / 0.93	87 / 1.07	\$7,662	26 / 0.82	-\$7,939
n=30, z=2.5, cls=low, mode=reversal	both	479 / 1.06	318 / 1.07	189 / 0.63	-\$89,556	46 / 0.82	-\$15,069
n=60, z=2.0, cls=low, mode=reversal	long	555 / 0.94	346 / 1.05	225 / 1.19	\$44,350	42 / 1.11	\$7,452
n=60, z=2.0, cls=low, mode=reversal	short	331 / 1.34	206 / 1.21	124 / 0.81	-\$31,191	37 / 0.86	-\$8,708
n=60, z=2.0, cls=low, mode=reversal	both	684 / 1.00	405 / 1.20	276 / 0.82	-\$54,479	60 / 1.25	\$21,775
n=60, z=2.5, cls=low, mode=reversal	long	241 / 0.88	192 / 1.07	85 / 1.00	-\$85	20 / 0.82	-\$6,740
n=60, z=2.5, cls=low, mode=reversal	short	144 / 1.00	91 / 1.46	59 / 0.83	-\$15,496	14 / 1.70	\$16,164
n=60, z=2.5, cls=low, mode=reversal	both	309 / 0.94	217 / 1.23	118 / 0.67	-\$55,027	26 / 1.34	\$15,726

Neighbours with ≥ 20 trades and positive net: Discovery 24/48; Confirmation 33/48; Holdout 26/48; 2026 20/45

Stop / exit management. The frozen entry under every management variant from the campaign library (discovery stages A and B).

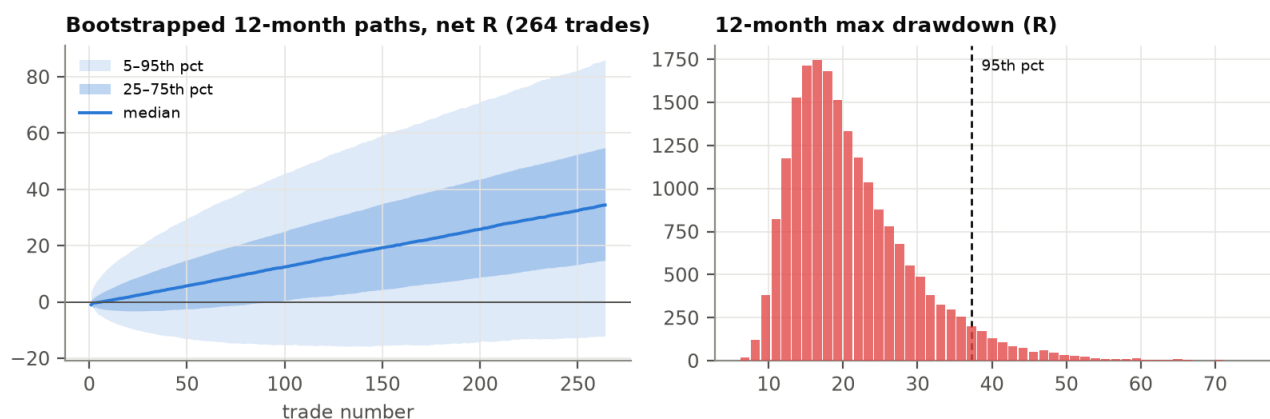
Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; single target at +1R	0.95	1.13	1332 / 1.07	\$43,212	0.79	-\$45,797
1×ATR stop; single target at +1.5R	0.98	1.15	1222 / 1.09	\$53,467	0.84	-\$38,237
1×ATR stop; single target at +2R	1.01	1.12	1142 / 1.07	\$43,832	0.80	-\$47,277
1×ATR stop; single target at +3R	1.08	1.20	1055 / 1.10	\$61,055	0.89	-\$23,752
1×ATR stop; single target at +4R	1.09	1.19	1014 / 1.06	\$38,044	0.80	-\$45,924
1×ATR stop; no target, flat 16:00	1.19	1.21	900 / 1.23	\$125,965	0.80	-\$40,214
1×ATR stop; no target, exit after 60 min	0.97	1.13	1130 / 1.18	\$94,245	0.98	-\$3,211
1.5×ATR stop; single target at +1R	0.99	1.18	1142 / 1.09	\$60,362	0.80	-\$50,307
1.5×ATR stop; single target at +2R	1.07	1.24	961 / 1.12	\$77,111	0.88	-\$29,787
1.5×ATR stop; no target, flat 16:00 ◀ frozen	1.17	1.26	795 / 1.21	\$120,595	0.77	-\$52,593
5-min swing stop; single target at +1R	1.08	1.14	887 / 1.14	\$97,582	1.00	\$813
5-min swing stop; single target at +2R	1.07	1.19	754 / 1.10	\$63,244	0.95	-\$11,769
5-min swing stop; no target, flat 16:00	1.13	1.18	689 / 1.19	\$112,329	0.97	-\$4,920
1×ATR stop; stop to entry at +0.5R, target +2R	0.71	1.00	1339 / 0.83	-\$68,776	0.50	-\$81,527
1×ATR stop; stop to entry at +0.5R, no target	0.81	1.02	1239 / 0.87	-\$48,616	0.50	-\$78,376
1×ATR stop; stop to entry at +1R, target +2R	0.91	1.15	1197 / 1.05	\$27,057	0.77	-\$45,086
1×ATR stop; stop to entry at +1R, no target	1.07	1.21	1039 / 1.14	\$64,439	0.70	-\$52,157
1×ATR stop; stop to entry at +1.5R, target +2R	0.98	1.15	1157 / 1.09	\$53,932	0.84	-\$33,945
1×ATR stop; stop to entry at +1.5R, no target	1.16	1.19	961 / 1.22	\$106,621	0.92	-\$14,019
1×ATR stop; no target, 1.5×ATR trailing stop	1.05	1.23	1100 / 1.08	\$42,735	0.83	-\$33,658
1×ATR stop; no target, 2×ATR trailing stop	1.07	1.25	1021 / 1.15	\$80,301	0.89	-\$22,008
1×ATR stop; no target, 3×ATR trailing stop	1.14	1.24	933 / 1.23	\$121,693	0.84	-\$33,405
1×ATR stop; no target, trail beyond last 5-min swing	1.10	1.15	994 / 1.11	\$55,264	0.87	-\$23,463
1×ATR stop; 50% at +1R, 50% at +2R	0.98	1.10	1142 / 1.09	\$44,154	0.75	-\$48,840
1×ATR stop; 50% at +1R, 50% at +3R	1.02	1.18	1055 / 1.11	\$51,755	0.79	-\$39,202
1×ATR stop; 50% at +1R, rest trails 2×ATR	1.01	1.21	1019 / 1.15	\$67,522	0.78	-\$39,300
1×ATR stop; 25% at +1R, 75% runs to 16:00	1.16	1.22	900 / 1.23	\$106,144	0.76	-\$43,268
1×ATR stop; 50% at +1R then stop to entry, rest +3R	0.97	1.17	1137 / 1.06	\$27,882	0.78	-\$40,938
1×ATR stop; stop to entry at +1R, 2×ATR trail	0.98	1.24	1086 / 1.08	\$38,236	0.77	-\$41,604

Cost sensitivity. The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$39; Confirmation \$157; Holdout \$166; 2026 n/a (gross loss). The baseline assumption is \$14.

H20 (#1, long + short): risk, sizing and forward expectations



Bootstrap of 2007 trades from 2019-Aug 2026 (all out-of-sample relative to discovery), 264 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown.

Forward 12-month statistic (bootstrap)	In R	1 NQ at 2026 risk (1R ≈ \$2,275)	1 MNQ (1R ≈ \$227)
Median 12-month result	+34.4	\$78,340	\$7,834
5th percentile (bad year)	-12.0	-\$27,237	-\$2,724
95th percentile (good year)	+86.1	\$195,820	\$19,582
Probability of a losing year	12%		
Median max drawdown	19.3	\$43,885	\$4,389
95th pct max drawdown	37.3	\$84,821	\$8,482
99th pct max drawdown	48.2	\$109,560	\$10,956
Historical max drawdown 2010-26	58.5		

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

Portfolio context (25-strategy frozen cohort, 1 contract each)

Period	Corr. with rest of cohort (daily)	Median pairwise corr.	Most correlated members	Share of cohort P&L
holdout	0.64	0.14	#19 H41 both (0.64), #12 H43 both (0.50), #15 H41 short (0.38), #3 H09 short (0.35)	19%
2026	0.53	0.20	#19 H41 both (0.51), #11 H07 long (0.35), #7 H07 both (0.34), #2 H45 both (0.32)	-53%

Discussion

Strength of evidence. Over 2010–2026 the strategy made +393R on 4,103 trades (+0.096R per trade, t 3.33 in R, PF 1.16 in dollars). Its record after the freeze, 962 trades over 2023–Aug 2026, is +145R (+0.15R per trade, t 2.34). This is the only H20 evidence that could not have influenced its selection, and it is still clearly positive. The holdout p against no-edge worlds was 0.059, marginal on its own, but the member belongs to a cohort that beat its null at p 0.03.

How to read 2026. The $-\$52.6k$ is real money on one contract. But most of the gap between $-\$52.6k$ and $-4.3R$ is position size: at 2026 volatility one full contract risked about \$2,275 per trade, about 2.2 $\times$ the 2019–22 average. The 21R drawdown is inside the historical range: 28R in the holdout, 58R in discovery. What is new is that the weakness hit every nearby variant at once, which is what a regime change looks like early on. It is also what an ordinary bad stretch looks like. 7.5 months cannot tell the two apart, so the forward record has to.

Where the P&L comes from. About 53% of trades stop out (average $-1.05R$). The other 47% ride to 16:00 (average $+1.38R$). The win rate is about 38%, with profit coming from a minority of large trend days. Entries in the first half-hour (09:30–09:59) are the most productive ($+0.36R$ on average), and entries on the quietest third of days did best ($+0.26R$ overall, $+0.53R$ after 2022). After 2022 the short side ($+0.20R$) outperformed the long side ($+0.10R$). All of these are descriptive; none should be used to change the rules.

Execution realism. The strategy trades market orders on the most liquid equity-index future during RTH. Break-even cost is far above the \$14 baseline, and the stress case (2 ticks per side plus 1.5 $\times$ commission) still leaves the holdout at $+\$111.1k$, PF 1.19. Capacity is not a constraint at single-desk size.

Portfolio fit. Holdout daily correlation with the other 24 cohort members combined was 0.64 (0.64 with H41 momentum #19, 0.50 with H43 #12). If the desk trades H41/H43, H20 is largely the same momentum exposure and should share one momentum risk budget, not get its own.

Recommendation

Recommendation: CONDITIONAL. Do not allocate full size now; start a small, fixed-risk pilot alongside the true-forward record. This is the best-evidenced strategy in the campaign. It passed every preregistered stage including the untouched 2023–25 holdout ($+\$120.6k$, PF 1.21, every year positive). But it lost $-\$52.6k$ in 2026, and it is the cohort's most correlated member. Scale up only after the forward gate on the last paper is met.

Decision requested: approve H20 for a **pilot**, not a full allocation.

Pilot design. Trade MNQ (micro) contracts, sized so each trade risks a fixed dollar amount (contracts = risk budget $\div$ (1.5 $\times$ ATRh $\times$ \$2)), rounded down, with zero contracts if below one. Set the risk per trade so the bootstrap 99th-percentile 12-month drawdown (48R) is $\leq 5\%$ of the capital assigned. Example: \$500 per trade means a \$24k 99%-case drawdown, so at least \$480k assigned. The pilot runs alongside the protocol's true-forward paper record from 24 Sep 2026, and the forward record is what decides scale-up.

Alternative if the committee wants no live exposure: forward-paper only, with the same gate below. At about 260 trades a year the gate is reached in about 7 months.

Conditions, limits and kill criteria

- Scale-up gate: at least 150 true-forward trades with average $\geq +0.08R$, net positive after stress costs, and no drawdown beyond 37R (the bootstrap 95th percentile). Then risk per trade may double. Full size needs 12 months forward at $\geq +0.08R$.
- Kill switch (stop the pilot and escalate): drawdown beyond 48R (bootstrap 99th percentile, above everything seen since 2019), or a rolling-150-trade average below $-0.05R$.
- Regime watch: if the 2026 pattern persists (continuation variants all PF < 1 while reversal variants are positive), treat it as evidence the efficient-momentum behaviour has changed, and stop even if the kill switch has not triggered.
- Risk budget: size H20 together with the H41/H43 momentum members as one momentum sleeve, since daily correlation is 0.5–0.64.

- No parameter, filter or management changes. Every observation in this report is descriptive. Changes would be a new hypothesis, preregistered and tested on future data only.

Caveats

- Backtests use 1-minute bars with pessimistic same-bar handling and fixed slippage. They do not model queue position, fast-market slippage beyond 2 ticks per side (stress case), or exchange outages.
- The 2026 sample (Jan–10 Aug) is short. The campaign's own record rates it low-power (cohort null p 0.26).
- The R-based view and the parameter/management grids in this report were computed after the holdout and 2026 results were known. They are diagnostic only and must not be used to modify or re-select the strategy. The preregistered process says the only valid new evidence now comes from the true-forward record starting 24 Sep 2026.
- This is a quantitative research assessment of a systematic strategy, not personalised investment advice. Capital allocation decisions rest with the committee.

Appendix A: frozen specification (verbatim)

H20 (#1, long + short)

```
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    "baseline_round_trip_usd": 14.0,
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  "entry_conditions": {
    "down_event": "the 30-minute move first falls below -2.0 sigma units with path efficiency x sqrt(n) >= 2.4",
    "mode": "continuation",
    "rule": "continuation: BUY on up-event, SELL on down-event; direction=both",
    "up_event": "the 30-minute move first exceeds +2.0 sigma units with path efficiency x sqrt(n) >= 2.4"
  },
  "entry_id": "Eac20f7491875358c",
  "family": "v3_h20_extreme_move_path",
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  "trading_window": {
    "entry_end": "15:30",
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    "name": "rth"
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}
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Appendix B: provenance

Item	Value
Final cohort freeze	freezes/FINAL_COHORT_FREEZE.json · sha256 f5666dac728aa2539b2f8223309012914d71923ab71f3363e9e035344ace8f03
Frozen (UTC)	2026-09-22T23:21:51.429326+00:00
Discovery freeze	sha256 0343a974953d53dc026fd1f7ef5c124c8985213b132fba5df4f8a986fc85b2c9
Master record	results/V3_MASTER_RESEARCH_RECORD.md
Sealed holdout / audit trades	results/holdout/trades.parquet · results/audit2026/trades.parquet
This report	reports/H20_Extreme_Move_Path_Research_Report.pdf, built from trades regenerated with the frozen specs (exact match with the sealed records) + stored discovery/confirmation evidence
Known erratum	results/audit2026/ERRATUM_2026-09-22.json (X3 null p only; does not affect this report)